Jihwan Gong

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Education

Seoul National University Seoul, South Korea
B.S. in Electrical and Computer Engineering; GPA: 4.06/4.30 (97.60%) *Mar. 2021 – Feb. 2027 (Expected)*

- **Relevant Coursework:** Deep Learning, Machine Learning Fundamentals, Mathematical Foundations of Deep Neural Networks, Introduction to Intelligent Systems, Introduction to Robotics

Research Experience

Undergraduate Researcher, Robot Learning & VLA Group Seoul, South Korea
Information Systems Laboratory (ISLAB), advised by Prof. Byonghyo Shim, Seoul National University *Dec. 2025 – Present*

- Co-authoring (**2nd author**) a manuscript on active perception in cluttered scenes, in preparation for **CoRL 2026**; responsible for implementing and benchmarking baseline methods and conducting real-world robot experiments.
- Completed directed study of the modern AI stack – from deep learning foundations through LLMs, VLMs, and Vision-Language-Action (VLA) models – with broader exploration of robot learning topics including humanoid manipulation.

Undergraduate Research Intern Seoul, South Korea
Dynamic Robotic Systems Laboratory (DYROS), advised by Prof. Jaeheung Park, Seoul National University *Jul. 2025 – Aug. 2025*

- Studied foundations of reinforcement learning through paper-reading and code reproduction.
- Participated in experiments on TOCABI humanoid and Franka manipulator platforms.

Publications & Preprints

[P1] First Author, Jihwan Gong, et al. *Learning Decomposed Visibility for Active Exploration of Cluttered Scenes. In preparation, target: CoR*

Selected Projects

Autonomous Racing on F1Tenth-style Platform | *ROS2, PyTorch, PPO, Diffusion Policy* [code]

- Designed and trained autonomous racing policies for 1/10-scale physical RC cars on an F1Tenth-inspired competition track (SNU *Introduction to Intelligent Systems*, Fall 2025).
- Implemented PPO baseline and additionally explored Diffusion Policy as an imitation learning alternative for high-speed racing control.

Humanoid Robot System for 2026 Humanoid Challenge | *SHAPE Humanoid Robotics Club, SNU* [code]

- Building components for SHAPE club's entry to the **2026 Humanoid Challenge** (Jul. 2026); prototyped an initial toy reasoning system as preparation.

Coverage Path Planning for 6-DoF Manipulator | *ROS2, AgileX Piper, Sigma Intelligence Club*

- Designed and implemented a coverage path planning (CPP) algorithm enabling a Piper manipulator to fully cover (paint) a designated 2D surface area (2025).

Voice-Controlled Beverage Serving Robot | *ROBOTIS Manipulator, YOLO, Google Speech-to-Text*

- Built an end-to-end serving robot integrating YOLO-based object detection and Google STT; system recognized 4 custom-trained beverage classes via natural-language voice commands (Sigma Intelligence Club, 2022).

Honors & Awards

Woonhae Foundation Scholarship, Woonhae Scholarship Foundation 2026
Honorable Mention, Hyundai Mobis Mobility Software Hackathon 2023
Department Scholarship, SNU Department of Electrical and Computer Engineering *(multiple semesters)* 2022 – 2026
Hansung Scholarship, Hansung Sonjaehan Scholarship Foundation *(high school)* 2019

Military Service

Republic of Korea Air Force, Mandatory Military Service Apr. 2023 – Jan. 2025

Technical Skills

Languages: Python, C++, C, MATLAB
ML & Robotics: PyTorch, NumPy, ROS2, MuJoCo, Isaac Gym & Isaac Lab
Tools: Git, Docker, Linux